Fixed testing functions for the strong contact branch

The complete state quadratic form can be completed uniformly when the native contact parameter has a fixed positive lower bound. The construction below gives its six by six Gram matrix by polynomial division, then assigns source-only positive matrices to fixed testing functions. It complements the weak-contact construction in *Completing the state quadratic form at a singular contact*. All source and testing variables are independent.

Native source. Throughout this note H, z, y, r, t mean the original native source parameters, rather than the compact variables used for the weak branch. Put

$$\varepsilon = 2^{-64}, \quad 2^{-12} \le H < 1, \quad 0 < z, y \le \varepsilon H, \quad 0 < r < 1, \quad t > 0,$$
$$a_c = 1 + zr, \quad j = a_c - t > 0, \quad B = H + yt, \quad S = a_c + yt.$$

Define

$$\begin{aligned}
\Delta &= z + (1+z)B + (1+z)B^2 + 2z(1+z)Br + z^3r^2 + 2z^2r + y\{(1+z)B + z + z^2r(2-r)\},\\
G_s &= B + z(H + r - 1), \qquad M_s = (1+z)(B+zr)(H+y) - z(1+y)(1-r)yt, \qquad (1)\\
n &= \frac{(B+zr)a_cS - r\Delta}{2r\Delta} \ge 0, \qquad m = \frac{z^2(1+y)(1-r)a_cS}{2G_s\Delta},\\
C &= \frac{a_cM_s}{2yt\Delta}, \qquad A = \frac{ya_c(B+zr)S}{2H\Delta}, \qquad D = \frac{(1+z)(B+zr)(B-zyr)S}{2yj\Delta}, \qquad \alpha = \frac{(B+zr)M_s}{2(1-H)\Delta}. \qquad (2)
\end{aligned}$$

The condition $n \ge 0$ is the physical source restriction. We normalize $d = 1, \lambda = \beta = 0$. For independent real testing angles and state coordinates set

$$D_u = 1 + u^2, \quad D_v = 1 + v^2, \quad J = uW + uvX - Y - vZ, \qquad M = vW - X + uvY - uZ$$

and write the complete, unnormalized slack as

$$\begin{aligned}
F = {} & 4nD_uD_vX^2 + 4mM^2 + 4CD_v(uW - Y)^2 + 4AD_uD_vY^2\\
& + 4DJ^2 + 4\alpha D_u(vY - Z)^2 + 2\{J^2 + D_u(vY - Z)^2 + D_v(X + uZ)^2 - D_uD_vZ^2\}.
\end{aligned} \qquad (3)$$

Its compact value is $S_{\mathbf{c}} = F/(4D_uD_v)$.

Theorem 1. *Every source* (2) *in the displayed domain has a Gram representation of $S_{\mathbf{c}}$, for all real state coordinates and*

$$|u| \le 2^{-12}, \qquad |v| \le 2^{-25},$$

in the following fixed testing family. Its base consists of the twenty functions $\mathcal{H}_{20}/\sqrt{D_uD_v}$ specified below. For each of the thirteen nonconstant exponents in Table 1, append the two eleven-function blocks

$$\frac{\sqrt{1 \pm (4096u)^i(33554432v)^j}}{\sqrt{D_uD_v}}\,\mathcal{H}_{11}.$$

All Gram entries depend only on the source. The representation includes literal $n = 0$ and arbitrary nonnegative additions of the eight event terms.

1 The complete remainder and its polynomial Gram

Let $k = \sqrt{y}$ and define the pure-source quantities

$$\begin{aligned}
&s = a_c/B, \quad c_* = Cyt, \quad d_* = Dyj, \quad \ell = y(C+D), \quad q = D/(C+D), \quad g = yCD/(C+D),\\
&\ell_* = H/4, \quad \alpha_* = H^2/8, \quad \mu = m/z^2, \quad \nu = nr, \quad a_\alpha = \alpha(1-H), \quad a_A = A/y.
\end{aligned} \qquad (4)$$

The testing rows are

$$\begin{aligned}
&\xi = X + zrvW/B, && \eta = Y - HuW/B, && \zeta = Z - uvW/B,\\
&R_h = (uvX - vZ + syuW)/k, && N = \{Y - quvX + qvZ - (H/B + qsy)uW\}/k, &&\\
&R = v(u^2 - z)W, && T = uk(u^2 - z)W. && \qquad (5)
\end{aligned}$$

In particular $\eta = k(N + qR_h)$. The four terms

$$\mathcal{R}_+ = 4(\ell - \ell_*)N^2 + 4Cv^2\eta^2 + 4nD_uD_v\xi^2 + 4(\alpha - \alpha_*)D_u(v\eta - \zeta)^2 \tag{6}$$

remain whole. Their coefficients are nonnegative by Section 3.

Here is a polynomial prescription for every entry of the remaining Gram. In the following expression regard N, ξ, ζ, R_h, W as indeterminates, and abbreviate

$$X_0 = \xi - zrvW/B, \quad Y_0 = k(N + qR_h) + HuW/B, \quad Z_0 = \zeta + uvW/B, \quad K = ukW,$$

$$J_0 = uW - Y_0 + v(uX_0 - Z_0), \qquad M_0 = vW - X_0 + uvY_0 - uZ_0.$$

The complete remainder is

$$\begin{aligned}
\mathcal{G} = {} & 4\ell_*N^2 + 4gR_h^2 + \frac{8(d_* - c_*)}{B}KN - 8gsKR_h + \frac{4(c_*t + d_*j)}{B^2}K^2 \\
& + \frac{4c_*t}{B^2}v^2K^2 - \frac{8c_*}{B}v^2K(N + qR_h) \\
& + 4\nu D_uD_v\left(\frac{z^2rv^2}{B^2}W^2 - \frac{2zv}{B}W\xi\right) \\
& + 4\alpha_*D_u(vk(N + qR_h) - \zeta)^2 \\
& + 4a_\alpha D_u\left\{\frac{(1 - H)u^2v^2}{B^2}W^2 - \frac{2uv}{B}W(vk(N + qR_h) - \zeta)\right\} \\
& + 4z^2\mu M_0^2 + 4ya_AD_uD_vY_0^2 + 2\{J_0^2 + D_u(vY_0 - Z_0)^2 + D_v(X_0 + uZ_0)^2 - D_uD_vZ_0^2\}.
\end{aligned} \tag{7}$$

Expanding the event squares gives $F = \mathcal{G} + \mathcal{R}_+$ at (5). Thus both state normals are still present.

Put $U = u^2, V = v^2$. Coefficients below refer to the five state indeterminates in $\mathcal{G}$. Define

$$\begin{aligned}
\rho &= \frac{[R_hW]\mathcal{G}}{uk}, & p_\xi &= \frac{[\xi W]\mathcal{G}}{v} + u^2\rho, & p_\zeta &= \frac{[\zeta W]\mathcal{G}}{uv} - \rho, \\
p_N &= \frac{[NW]\mathcal{G}}{uk}, & \phi &= [W^2]\mathcal{G} - su^2(v^2 - y)\rho.
\end{aligned} \tag{8}$$

Each division in (8) is exact polynomial cancellation; the resulting four expressions $p_\xi, p_\zeta, p_N, \phi$ are polynomials in U, V. They define their values also on $u = 0$ or $v = 0$. Take the following monic divisions, which uniquely define all quotients and remainders:

$$\begin{aligned}
p_\xi &= (U - z)A_\xi + z\{(V - y)B_\xi + r_\xi\}, \\
p_\zeta &= (U - z)A_\zeta + (V - y)B_\zeta + r_\zeta, \qquad p_N = (U - z)A_N + (V - y)B_N + r_N, \\
d_{\mathrm{rad}} &= \frac{4}{B^2}\{z^2\mu(B + zr)^2 + (\nu r + r^2/2)z^2\}, \quad e_{\mathrm{rad}} = \frac{4}{B^2}(\nu r + r^2/2)z^2,
\end{aligned} \tag{9}$$

$$\begin{aligned}
\phi - \frac{d_{\mathrm{rad}} + e_{\mathrm{rad}}V}{z^2}V(U - z)^2 &= a_2(V)U^2 + a_1(V)U, \\
a_2 = (V - y)A_s + r_a, \qquad a_1 &= (V - y)B_s + r_b, \qquad B_s + zA_s = (V - y)C_s + r_c.
\end{aligned} \tag{10}$$

The symbol B_s in the last line is a quotient polynomial, unrelated to the native scalar B. No division by a testing value is used.

To explain why the six remainders vanish, the native stationary contact is

$$u = \sqrt{z}, \quad v = \sqrt{y}, \quad (W, X, Y, Z) = \left(B/\sqrt{zy}, -r\sqrt{z}, H/\sqrt{y}, 1\right).$$

At this point $N = \xi = \zeta = R_h = R = T = 0$. The rational weights (2) give zero parent value and zero first derivatives in the three independent normal directions and in u, v. After the ρ-elimination, the three normal derivatives give $p_\xi(z, y) = p_\zeta(z, y) = p_N(z, y) = 0$. The value and two angle derivatives give $\phi(z, y) = \phi_U(z, y) = \phi_V(z, y) = 0$. Equations (9)–(10) then give

$$r_\xi = r_\zeta = r_N = r_a = r_b = r_c = 0. \tag{11}$$

Indeed the scalar value and derivatives are respectively $z^2r_a + zr_b$, $2zr_a + r_b$ and zr_c; since $z > 0$, their vanishing determines all three scalar remainders. These are exact source identities: equivalently, substitute (2) in the six explicitly defined remainders and clear their positive source denominators.

Let $\mathcal{G}_0$ be the part of $\mathcal{G}$ quadratic in N,ξ,ζ,R_h, with no W. In the next formula evaluate every quotient at $U=u^2, V=v^2$, and put $Q_n=u\xi-\zeta$:

$$\begin{aligned}\mathcal{O}=\mathcal{G}_0&+\{A_\xi-(v^2-y)B_\xi\}R\xi+\frac{B_\xi}{s}(uv^2Q_n-uvkR_h)\xi\\&+uA_\zeta R\zeta+\frac{B_\zeta}{s}(v^2Q_n-vkR_h)\zeta+A_NTN+\frac{B_N}{s}(vkQ_n-yR_h)N\\&+\frac{d_{\mathrm{rad}}+e_{\mathrm{rad}}v^2}{z^2}R^2+\frac{A_s}{s}(-R_hT+uRQ_n)+\frac{C_s}{s^2}(v^2Q_n^2-2vkR_hQ_n+yR_h^2).\end{aligned}\tag{12}$$

This explicitly defines the symmetric six by six matrix

$$f=(N,\xi,\zeta,R_h,R,T)^T,\qquad Q=\tfrac12\nabla_f^2\mathcal{O},\qquad \mathcal{O}=f^TQf.$$

Indeed, using $kR_h=vQ_n+su(y-v^2)W$, direct monic division gives

$$\mathcal{G}-\mathcal{O}=vzr_\xi\xi W+uvr_\zeta\zeta W+ukr_NNW+\{u^4r_a+u^2r_b+u^2(v^2-y)r_c\}W^2.\tag{13}$$

Together with (11), this proves the entire parent identity. In particular d_*,t,j occur only in the vanishing remainders and need no interval replacement in Q.

2 Removing the degenerate contact directions

The exact testing relation

$$\mathcal{C}=T^2-u^2R^2-\frac{u^2-z}{s}R_hT+\frac{u(u^2-z)}{s}R(u\xi-\zeta)=0\tag{14}$$

follows at once from (5). Add $4\mathcal{C}/H$ to f^TQf, and use

$$\widehat{f}=\left(\sqrt{H}N,\ \xi+u\zeta+\frac{R}{H(1+H)},\ H\zeta,\ \sqrt{H}R_h,\ R/\sqrt{H},\ T/\sqrt{H}\right)^T.\tag{15}$$

If $h=\sqrt{H}$, the matrix taking $\widehat{f}$ to f has diagonal $(h^{-1},1,H^{-1},h^{-1},h,h)$ and additional entries $L_{23}=-u/H,\ L_{25}=-1/[h(1+H)]$. Thus the exact normalized matrix is

$$\widehat{Q}=L^T\left(Q+\frac{4}{H}\,\frac{\nabla_f^2\mathcal{C}}{2}\right)L.\tag{16}$$

Its coefficients are obtained from the displayed polynomial operations; no entries are chosen by a semidefinite feasibility problem.

The leading source matrix is

$$Q_*=\operatorname{diag}\left(1,2,\tfrac12,\frac{2}{1+H},\frac{2}{(1+H)^2},4\right)+\frac{e_4e_6^T+e_6e_4^T}{1+H}\succeq\tfrac12 I_6.\tag{17}$$

For the last inequality, the fifth diagonal after subtraction of $I_6/2$ is $(1-H)(3+H)/[2(1+H)^2]$. The fourth/sixth block has positive first pivot and determinant $[17+7H(2-H)]/[4(1+H)^2]$; the other entries are nonnegative.

3 Uniform source and coefficient bounds

Set $a_0=1/[2(1+H)]$. The nine errors are defined by

$$\begin{array}{lll}B/H=1+e_B, & sH=1+e_s, & c_*/H=a_0+e_C,\\ g/H=a_0+e_g, & a_\alpha/H^2=a_0+e_\alpha, & AH/y=a_0+e_A,\\ mH^2/z^2=(1-r)a_0+e_m, & nr=a_0-r/2+e_n, & q=t+e_q.\end{array}\tag{18}$$

They satisfy

$$|e_B|\le 2\varepsilon,\quad |e_s|\le 4\varepsilon,\quad |e_C|,|e_\alpha|,|e_A|,|e_m|,|e_n|\le 16\varepsilon,\quad |e_g|,|e_q|\le 64\varepsilon.\tag{19}$$

Here is a factorwise proof uniform in $r,t,j,1-H$. Writing $T_s=B+zr$ and $U_s=B-zyr$, the source formulas give

$$\begin{gathered}1\le T_s/H,\ S\le(1+\varepsilon)^2,\qquad 1-\varepsilon^2\le U_s/H\le(1+\varepsilon)^2,\\ 1-\varepsilon\le G_s/H\le 1+3\varepsilon,\quad 1\le M_s/H^2\le(1+\varepsilon)^4,\quad 1\le\Delta/[H(1+H)]\le 1+10\varepsilon.\end{gathered}\tag{20}$$

The lower bound for M_s uses

$$M_s = (1+z)T_sH + yG_s + yzj + zr(yt)(1+y);$$

its upper bound uses its first expression in (1) and drops the nonnegative subtracted term. The other elementary bounds follow from $t < a_c \le 1+\varepsilon$. For completeness, $B \le q_0 H$ with $q_0 = 1+\varepsilon+\varepsilon^2$. Termwise bounds in Δ give

$$\Delta - H(1+H) \le H(a + bH + cH^2), \quad a = 2\varepsilon + \varepsilon^2 \le 3\varepsilon,$$

$$b = 6\varepsilon + 10\varepsilon^2 + 6\varepsilon^3 + \varepsilon^4 \le 7\varepsilon, \quad c = \varepsilon + 5\varepsilon^2 + 8\varepsilon^3 + 5\varepsilon^4 + \varepsilon^5 \le 2\varepsilon.$$

Since $3 + 7H + 2H^2 \le 6(1+H)$ for $0 \le H \le 1$, this even gives the stronger upper bound $1 + 6\varepsilon$.

The ratios of $c_*/H, d_*/H, a_\alpha/H^2, AH/y$ to a_0 have lower bound $(1-\varepsilon^2)/(1+10\varepsilon)$ and respective upper bounds $(1+\varepsilon)^5, (1+\varepsilon)^7, (1+\varepsilon)^6, (1+\varepsilon)^5$. Their relative errors are at most 12ε. The ratio of mH^2/z^2 to $(1-r)a_0$ lies between

$$\frac{1}{(1+3\varepsilon)(1+10\varepsilon)} \quad \text{and} \quad \frac{(1+\varepsilon)^4}{1-\varepsilon},$$

and $nr + r/2 = AH/y$ exactly. These imply the corresponding 16ε bounds. Also $|e_B| \le \varepsilon(1+\varepsilon)$ and $|e_s| \le 3\varepsilon$. If $c_*/H = a_0(1+\delta_C)$, $d_*/H = a_0(1+\delta_D)$, put $\delta_w = (j\delta_C + t\delta_D)/a_c$. Then $|\delta_w| \le 12\varepsilon$ and

$$q = \frac{t}{a_c}\frac{1+\delta_D}{1+\delta_w}, \qquad \frac{g}{H} = \frac{a_0}{a_c}\frac{(1+\delta_C)(1+\delta_D)}{1+\delta_w}.$$

The endpoint ratios give $|q - t| < 26\varepsilon$ and $|g/H - a_0| < 37\varepsilon/2$, proving (19). In particular

$$\ell \ge 4g \ge H(1 - 256\varepsilon) > H/4, \qquad \alpha \ge H^2(1/4 - 16\varepsilon) > H^2/8.$$

This proves positivity of every coefficient in (6).

The following finite majorant specifies the coefficient estimates without long rational-entry lists. In (16) substitute

$$H = h^2, \quad B = h^2(1+e_B), \quad s = (1+e_s)/h^2, \quad z = h^2 z_1, \quad k = hk_1,$$

and replace the other quantities by (18), with $\ell_* = h^2/4, \alpha_* = h^4/8$. Expand $\widehat{Q} = \sum C_{ij} u^i v^j$. Absorb any positive rational constant in a denominator into its numerator. Every remaining denominator factors into $h^a(1+h^2)^b(1+e_B)^c(1+e_s)^d$. For a numerator monomial with h-exponent a', bound its h factor by $64^{\max(a-a',0)}$, its remaining source factors using (19),

$$|r| \le 1, \quad |t| \le 2, \quad |z_1| \le \varepsilon, \quad |k_1| \le 2^{-32},$$

and its denominator by 2^{c+d}. Sum absolute coefficients and count symmetric off-diagonal entries twice. The triangle inequality bounds the operator norm by this explicitly defined finite sum. The result for $C_{00} - Q_*$, and for each $2^{-12i-25j}C_{ij}$, is bounded by Table 1. It uses only the divisions (9)–(10) and the two Hessians above; thus it also specifies every rational coefficient estimate. There are exactly fourteen monomials and fifty-one nonzero

(i,j)	00	01	02	10	11	12	20
α_{ij}	2^{-40}	2^{-50}	2^{-33}	2^{-4}	2^{-55}	2^{-41}	2^{-18}
(i,j)	21	22	30	32	40	42	50
α_{ij}	2^{-62}	2^{-48}	2^{-28}	2^{-65}	2^{-42}	2^{-72}	2^{-51}

Table 1: Fixed rational coefficient budgets. The constant entry bounds $\|C_{00} - Q_*\|_2$; all others bound $\|2^{-12i-25j}C_{ij}\|_2$.

upper-triangular coefficient records. Their full exact definition is

$$(i,j,a,b) \longmapsto [u^a v^b]\left\{L^T\left(\tfrac{1}{2}\nabla_f^2\mathcal{O} + \tfrac{4}{H}\,\tfrac{1}{2}\nabla_f^2\mathcal{C}\right)L\right\}_{ij},$$

after the displayed source substitution. Off-diagonal entries are actual symmetric entries.

4 Positive fixed Gram blocks and state restoration

The sum of all fourteen budgets is

$$\frac{295165937727343559809}{4722366482869645213696} < \frac{1}{8}.$$

For each nonconstant exponent set

$$p_{ij} = (4096u)^i (33554432v)^j, \qquad A_{ij} = 2^{-12i-25j} C_{ij}, \qquad Q_{\mathrm{res}} = C_{00} - \sum_{(i,j)\neq(0,0)} \alpha_{ij} I_6.$$

Then $Q_{\mathrm{res}} \succeq 3I_6/8$, $(\alpha_{ij} I_6 \pm A_{ij})/2 \succeq 0$, and the exact identity

$$\widehat{Q} = Q_{\mathrm{res}} + \sum_{(i,j)\neq(0,0)} \left\{ \frac{\alpha_{ij} I_6 + A_{ij}}{2}(1 + p_{ij}) + \frac{\alpha_{ij} I_6 - A_{ij}}{2}(1 - p_{ij}) \right\}$$

proves positivity on the closed testing rectangle.

All six roots (15) are linear combinations, with source-only coefficients, of

$$\mathcal{H}_{11} = (X, Y, Z, vW, uvW, uW, uvX, vZ, u^2 vW, u^3 W, uZ).$$

The retained terms (6) use this list together with

$$(uX, vX, v^2 W, uv^2 W, vY, uvY, W, uvZ, uY),$$

giving $\mathcal{H}_{20}$. For example the n-term uses $\xi, u\xi, v\xi, uv\xi$, and the last retained term uses $v\eta - \zeta, u(v\eta - \zeta)$. The source expansions are polynomial in the original state coordinates, so arbitrary state signs and zero state coordinates are included.

Push the six-dimensional positive matrices to these fixed spans, restore $\mathcal{R}_+$, divide the Gram coefficients by four and the testing functions by $\sqrt{D_u D_v}$. This gives precisely $S_{\mathbf{c}}$, proving Theorem 1. No formula divides by n, so $n = 0$ is literal. The event roots already belong to $\mathcal{H}_{20}$; their nonnegative addition preserves the representation.

Together with the weak-contact family and the two preserved remote testing matrices, this supplies the native source envelope with one fixed certificate of total PSD order 635, including nine source signs. The remote matrices provide the projective testing endpoints. Admission of every canonical source is a separate requirement; the present theorem establishes no equality with the full canonical cone.